

Popquiz Free_Fall

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:52pm

Quiz Instructions

Free-fall means no air resistance. $a = -10\text{m/s/s}$ about



Question 1 20 pts

a ball is tossed up with a speed of 20m/s . we neglect air resistance.

At the top of the trajectory , what is true

(up is +, down is -)



velocity is 20m/s , acceleration is -10m/s/s



velocity is 0, acceleration is -10m/s/s



velocity is 0, acceleration is 0



velocity is -20m/s , acceleration is 10ms/s



Can not be determined



Question 2 20 pts

A watermelon is dropped from the empire state building. Its in free-fall. It falls for about 9s. Its speed before it reaches the ground is about



50m/s



20m/s



90m/s



10m/s



Question 3 20 pts

An apple falls from a tree and hits the ground 5 meters (-5) below. The speed of the apple before it hits the ground is about ?

hint: use the 4th equation. $V^2 = V_0^2 + 2(-10)(\text{displacement})$. with $V_0=0$

☐

15m/s

☐

20m/s

☐

5m/s

☐

10m/s

☐

Question 4 20 pts

You are tossing a ball up with an initial velocity of + 20m/s

It reaches the maximum height after ($V_f=0$)

☐

0.5s

☐

1s

☐

6s

☐

5s

☐

2s

☐

Question 5 20 pts

Lucie wants to find the height of a tower on miami beach using the kinematics equation $y(t)$ for free-fall.

She dropped a bowling ball (from rest) from the top of the tower (no was was injured). Her friend has a high speed camera and computed the time for the free-fall to be about 10s. The height is about



- ☐ 500m
- ☐ 100m
- ☐ 300m
- ☐ 50m

Not saved

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